Deepan Chandrasekaran

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE student with hands-on experience shipping production-grade LLM and GenAI systems — RAG pipelines, vector databases (FAISS, ChromaDB), LLM integrations, and real-time speech pipelines. Skilled in Python, FastAPI, Docker, PyTorch, and Hugging Face. Open-source contributor with a national hackathon win.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++
LLMs & GenAI: OpenAI API, Gemini API, Ollama (Mistral), Hugging Face, Prompt Engineering, Fine-tuning
RAG & Vector Search: FAISS, ChromaDB, Sentence Transformers, BM25 hybrid retrieval, RAG pipeline design
Frameworks: FastAPI, PyTorch, Faster-Whisper, WebRTC VAD, React, TailwindCSS, Three.js, Spring Boot
DevOps & Cloud: Docker, Microservices, Git, Linux, AWS (basics), Vercel, NVIDIA A100/H100/T4

PROJECTS

LaRa – Real-Time AI Therapy Platform | Jan 2026 – Present | Python, FastAPI, Docker, Faster-Whisper, RAG, WebRTC VAD

- Production-grade GenAI system with LLM narrative generation, deterministic FSM decision engine, RAG-based long-term memory, and real-time speech pipeline; containerized with Docker.
- Won 3rd Place at RAMPAGE V26 — National Level Hackathon (KL-H).

ResumeAnalyse – RAG Resume Matching System | Dec 2025 – Feb 2026 | Sentence Transformers, ChromaDB, Ollama, Mistral, React, TypeScript

- End-to-end RAG pipeline: PDF chunking → Sentence Transformer embeddings → ChromaDB vector DB → semantic retrieval; achieved >90% job-resume relevance.
- Wired Mistral LLM via Ollama for context-grounded Q&A; prompt engineering experiments cut recruiter analysis time by 50%.

Search Wizard – Semantic Local Search Engine | 2025 | Python, FAISS, Gemini Embeddings, BM25, FastAPI, Electron

- Multimodal semantic search over 50K+ files using FAISS (768-dim) + BM25 hybrid retrieval; FastAPI backend with assistant mode powered by Gemini Flash LLM.
- Open-source cross-platform Electron app with real-time indexing, vector search, and RAG-style document Q&A.

ISL Sign Language Recognition – CNN Model | Aug – Nov 2025 | PyTorch, CNNs, NVIDIA A100/H100/T4

- CNN trained on 42,000 images — 99.8% validation accuracy; real-time inference at 45 FPS; scalable to 1M+ samples on HPC GPUs.

WORK EXPERIENCE

Website Designer at **Local Apparel Manufacturer** | 2023 | Freelance

- Engineered a full-stack shirt customization platform (React, Three.js) with 3D preview and order management backend; boosted sales 8-10%.

EDUCATION & CERTIFICATIONS

B.Tech CSE — KL University, Vijayawada | Expected April 2027 | CGPA: 9.15 / 10
Oracle Cloud Infrastructure 2025 – Generative AI Professional | Oracle University | Oct 2025 – Oct 2027
Oracle AI Vector Search Professional | Oracle University | Oct 2025 – Oct 2027

CODING PROFILES

LeetCode: leetcode.com/u/kl2300032731 **CodeChef:** codechef.com/users/klu2300032731